

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458739.8174 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.011
Transit depth (Rp/Rs)^2	2.72 +/- 0.36 [%]
Orbital Inclination (inc)	86.66 +/- 1.59
Airmass coefficient 1 (a1)	1.048 +/- 0.011
Airmass coefficient 2 (a2)	-0.0421 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#3 - [257, 24]
Optimal Method	PSF photometry
Transit Duration (day)	0.1282 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.165 ± 0.011 vs 0.1489 (deviation 1.46σ)
Duration fit vs published	0.1282 d vs 0.1304 d (deviation 0.58σ)
Mid-time O–C	5.7 min
Depth SNR	15.0
Residual scatter	1.02 %

Light curve

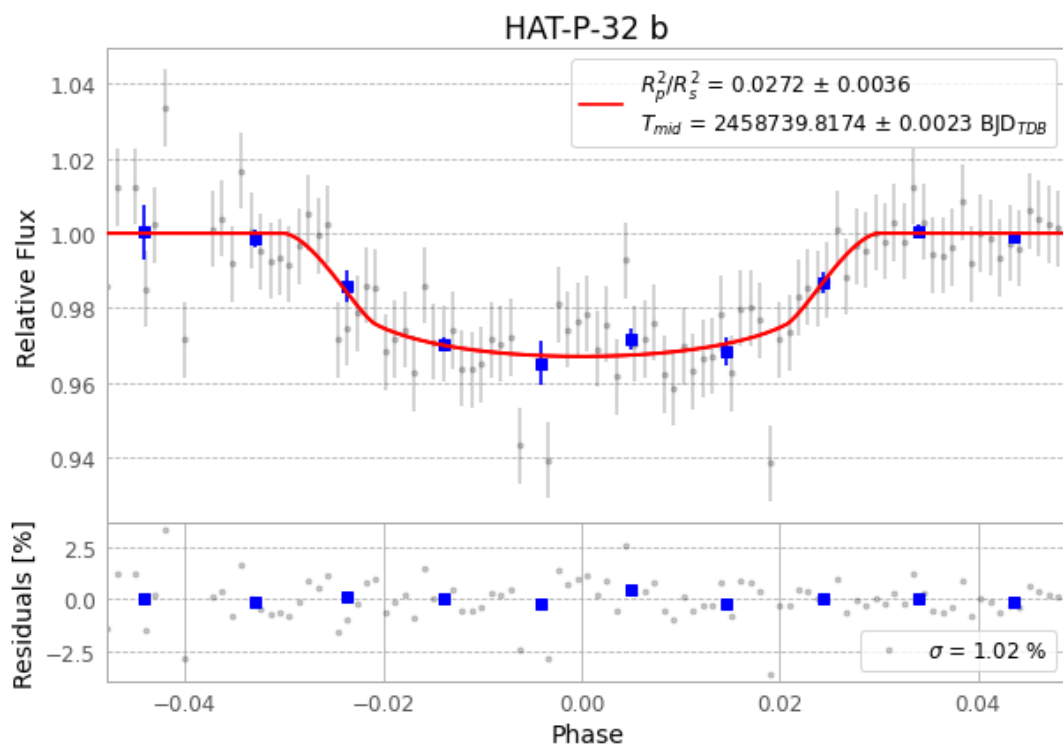


Figure 1 — FinalLightCurve_HAT-P-32 b_2019-09-12.png (SHA-256 c96fa8b11064bc82...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 95], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31fad0f7e82e6e80...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), HATP-32190913045712.FITS → HATP-32190913100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-190913.log (66cb41a1d0f9...), FinalLightCurve_HAT-P-32 b_2019-09-12.png (c96fa8b11064...), FinalLightCurve_HAT-P-32 b_2019-09-12.csv (d3376a6e4805...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:18Z.